Buckyball Colourings

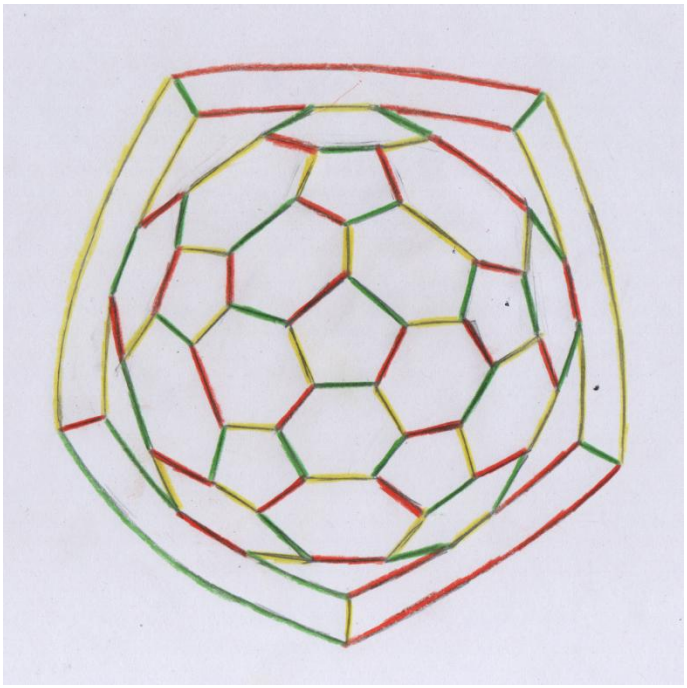


Figure 1. Three-colouring

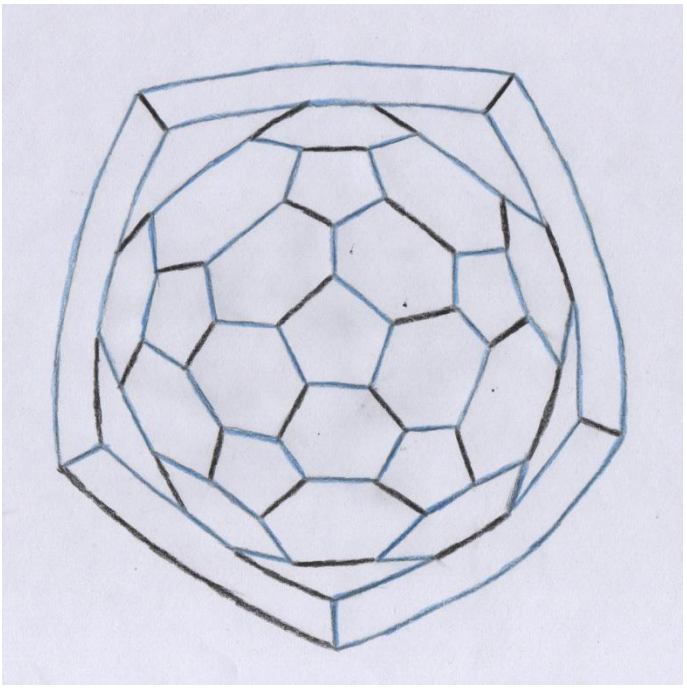


Figure 2. Hamiltonian cycle

Modular origami is one of the best “hands-on” ways to explore mathematics, and here are two of my favourite examples from the subject known as “graph theory”.

The “graph” of an object is a 2D representation that allows us to look at it from a different perspective. In this case, the object is known as a “buckyball” which is similar to a soccer ball, so think of making a soccer ball out of stretchy wire and flattening it against a wall.

Figure 1 is an example of what is known as a “three-colouring”, that is an assignment of three colours such that no two edges of the same colour touch. This is not possible for every graph, but I knew it was possible for the buckyball, so I used trial and error (and the help of a friend) to eventually come up with a solution. Interestingly, the colours are not “evenly” distributed, which makes this a very challenging problem (it took me over 4 hours!). For a more algorithmic approach, have a look at the following page by Thomas Hull:

http://origametry.net/papers/buckycolortalk1.pdf

Figure 2 illustrates a related concept known as a “Hamiltonian cycle” which is essentially a continuous loop (blue) through each vertex exactly once. This is a formidable problem that is well beyond the reach of trial and error, so I wrote a program in Python in order to find a solution. My program takes about 80 seconds to run which I believe is reasonable. It is important to note that a Hamiltonian cycle can be converted to a three-colouring (by alternating the colours on the cycle and choosing a third colour for the remaining edges), but not the other way around in general (my three-colouring does not generate a Hamiltonian cycle for instance).